

DMD: A DANISH SHIP DETECTION DATASET

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Abstract—The availability of high-quality, domain-specific datasets remains a limiting factor for maritime object detection research, particularly for monitoring heavily trafficked shipping routes under real-world conditions. We present the Danish Maritime Dataset (DMD), a publicly available image dataset collected from two fixed coastal observation points in Denmark: a camera mounted on the east pylon of the Great Belt Bridge and a camera located on the island of Sprogø. Both sensors continuously monitor the same Great Belt shipping route, one of the busiest maritime corridors in Northern Europe. The dataset captures maritime traffic under a wide range of conditions, including varying weather, illumination, and object scale. Moreover, a subset of the dataset provides simultaneous observations from the two view points. All images were systematically annotated by hand. As a baseline benchmark, we train and evaluate a YOLO-based object detection model on the dataset, providing reference performance metrics for future studies. By releasing both the dataset and benchmark results, this work aims to facilitate research on maritime vessel detection, long-term coastal monitoring, and robust object detection in real-world maritime environments.

Index Terms—Great Belt, Image Dataset, Camera, Maritime Domain Awareness, Object Detection, Multimodality.

I. INTRODUCTION

Maritime Domain Awareness (MDA) is a critical component of global trade safety and security. As the volume of international shipping traffic continues to rise, the ability to automatically monitor shipping lanes, detect anomalies, and classify vessels has become paramount. While the Automatic Identification Systems (AIS) provide foundational vessel data, it is susceptible to signal loss, jamming, spoofing, and the transponder may be turned off by accident or intentionally. Consequently, electro-optical sensors and computer vision have emerged as essential tools for robust object detection and tracking in maritime environments [1].

However, the efficacy of deep learning models in the maritime domain is inextricably linked to the quality and diversity of the training data. Unlike terrestrial domains, which benefit from massive, standardized benchmarks (e.g., COCO [2]), maritime environments present unique challenges. These include dynamic water surfaces, rapid illumination changes, severe weather conditions, and extreme scale variations [3].

The development of vision-based maritime datasets has gained significant traction in recent years to address these issues. Comprehensive surveys [1, 3] highlight the evolution of video processing in this domain, identifying a specific gap in datasets that capture continuous, long-term monitoring from varying elevations.

Early contributions, such as the ARGOS-Venice dataset [4], focused on specific, constrained environments. To address the need for larger-scale data, the SeaShips dataset [5] was introduced, comprising 7,000 images covering common ship types (e.g., ore carriers, container ships). Similarly, MCShips [6] was released as a large-scale dataset designed for fine-grained categorization, aggregating over 8,000 images. While these datasets are instrumental for general vessel recognition, they often lack the temporal consistency required for stationary surveillance applications or do not reflect the specific geometric challenges of high-elevation camera placement.

Aerial perspectives offer a different set of challenges, primarily related to top-down view angles and scale. Datasets specifically designed for airborne maritime surveillance [7] focus on detection from Unmanned Aerial Vehicles (UAVs). While useful for search-and-rescue operations, the visual features in top-down imagery often do not transfer well to the horizontal or slanted profiles seen by coastal surveillance cameras. Recent trends have also moved toward multi-sensor fusion, such as the M²SODAI dataset [8].

In this study, we introduce the Danish Maritime Dataset (DMD), a publicly available maritime image dataset designed to support research on fixed-infrastructure monitoring. The dataset targets long-term observation of dense maritime traffic in a real-world operational setting and captures substantial variability in environmental conditions, vessel appearance, and object scale. All images are manually annotated with bounding boxes to ensure high-quality ground truth. In addition to releasing the dataset, we provide baseline object detection results using a YOLO-based model to establish reference performance for future comparative studies.

II. DATA

The dataset consists of RGB images acquired from two fixed cameras monitoring the Great Belt Strait (Storebælt), Denmark. The strait is a major international shipping corridor connecting the Baltic Sea and the North Sea and is subject to continuous commercial traffic.

A. Sensors

Data were collected from two geographically separated, stationary vantage points that provide complementary views of the shipping lane.

The first camera is mounted on the eastern pylon of the Great Belt Bridge. It is installed at high elevation above sea level and provides a long-range, oblique view of the main

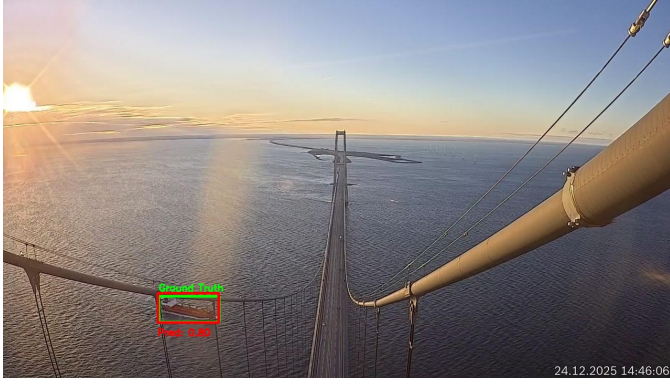


Fig. 1: Image from the East Pylon with sun glare.



Fig. 2: Image from Sprogø. Two ships are captured by the wide view angle.

traffic corridor. This configuration results in large variations in object scale and pronounced perspective effects, particularly for vessels at extended distances from the camera (see Fig. 1).

The second camera is located on the island of Sprogø, positioned closer to sea level relative to the bridge-mounted sensor. It captures the shipping lane from a horizontal viewing angle, yielding a side view of the vessels as they traverse the waterway, with very little change in scale (see Fig. 2).

B. Image Acquisition

Both cameras operate as fixed, static sensors and continuously monitor the waterway without active pan-tilt-zoom control. Image acquisition spans a wide range of environmental conditions, including variations in illumination, weather, sea state, and atmospheric visibility. The dataset includes vessels at different distances from the cameras, resulting in significant scale diversity across samples. The camera feeds are publicly available at <https://storebaelt.dk/trafik-vejtr/webcams/>.

All images are stored as standard RGB JPG files with a width of 1280 pixel and height of 720 pixel. The images are named by the acquisition UNIX timestamp e.g. *1765630219.jpg*. However, this timestamp can be slightly deviated from the image's actual recording time indicated by the watermark due to latency. No synthetic augmentation

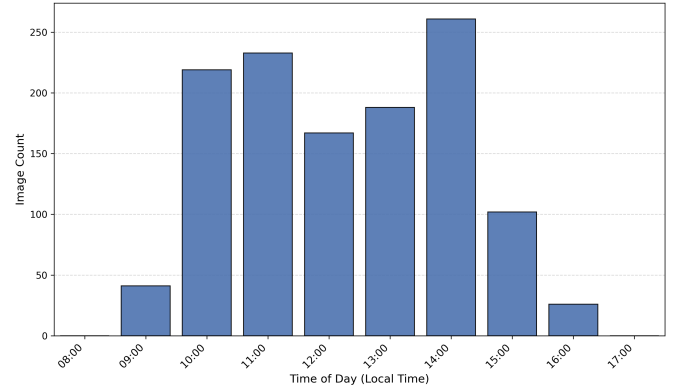


Fig. 3: Image distribution by hour of day

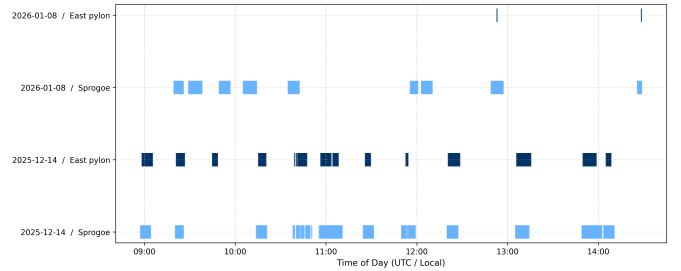


Fig. 4: Image co-views.

or simulated imagery is included in the dataset. All samples correspond to real-world observations.

The images were predominantly acquired during daytime (see Fig. 3) over the period from December 2025 to January 2026. Owing to the proximity of the acquisition period to the winter solstice and Denmark's northern latitude, local sunrise and sunset occur at approximately 08:30 and 16:30, respectively.

C. Annotation Procedure

Each image in the dataset was manually annotated with bounding boxes corresponding to visible maritime vessels. Annotations were produced through systematic human labeling to ensure consistency and accuracy across viewpoints and environmental conditions. The dataset is intended to support supervised object detection tasks and benchmark evaluation under fixed-infrastructure maritime surveillance settings.

D. Multimodality

Both cameras monitor the same high-traffic shipping lane, providing complementary views of vessels from different perspective. A subset of the dataset overlaps temporally, with both cameras simultaneously observing the waterway. In particular, the scenes from 14 December 2025 have the greatest overlap (see Fig. 4).

III. METHODOLOGY

The methodology of this study utilizes the YOLOv8 (You Only Look Once version 8) framework, the latest state-of-the-art iteration in the YOLO family. As a one-stage detector,

TABLE I: Pretraining evaluation

Pretraining	mAP50	Precision	Recall
-	0.86	0.94	0.75
SMD	0.91	0.92	0.85
SeaShips	0.92	0.99	0.82
SMD+SeaShips	0.93	0.89	0.92

YOLOv8 builds upon previous versions by implementing an anchor-free detection head and a decoupled structure that independently processes objectness, classification, and regression tasks [9]. These advancements allow for higher spatial precision and faster inference speeds compared to its predecessors. For this application, the YOLOv8n (nano) variant was selected to balance computational efficiency with detection accuracy. This model features a reduced number of parameters, making it ideal for providing baselines and benchmarks while maintaining a high Mean Average Precision (mAP) through its C2f (Cross Stage Partial Bottleneck with two convolutions) modules and optimized feature integration. The model was implemented using the Ultralytics library [10].

The model was trained on an NVIDIA GeForce RTX 4060 GPU (8GB VRAM). To address environmental complexities typical of coastal surveillance—such as specular sun glare and high-contrast shadows—a robust data augmentation strategy was implemented [11]. This included photometric distortions via Hue-Saturation-Value (HSV) shifts, geometric transformations involving random rotations ($\pm 5^\circ$) and scaling ($\pm 50\%$), and mosaic augmentation to improve context-aware detection. Fine-tuning was executed for 15 epochs at an increased input resolution of 1280×1280 pixels. This high-resolution configuration was specifically chosen to preserve the spatial features of small vessels at extended ranges, which are often obscured at lower pixel densities.

A. Experiments and benchmarks

To establish reference benchmarks and study the dataset in detail, we define a set of baseline experiments.

First, we pretrain a model on existing large-scale maritime datasets. Pretraining is performed using the Singapore Maritime Dataset (SMD)[1] and the SeaShips dataset[5]. The pretrained model is transferred to our dataset introduced in this work. To isolate the effect of pretraining, we compare detection performance against an identical model trained from random initialization. For this experiment, the images were randomly partitioned into 70% training, 20% validation and 10% test. The results are presented in Table I. It demonstrates the effectiveness of pretraining. All configurations using pre-training consistently achieved the best results. For subsequent experiments, we therefore adopt the model pretrained on the combined SMD and SeaShips datasets.

Table II presents the evaluation per camera. Comparable performance is observed across both observation sites, with slightly lower metrics obtained on the Sprogø data.

Next, cross-camera generalization is evaluated. Models are trained using data from one camera and tested on data from

TABLE II: Per-camera

Camera	mAP50	Precision	Recall
East Pylon	0.93	0.92	0.86
Sprogø	0.85	0.96	0.78

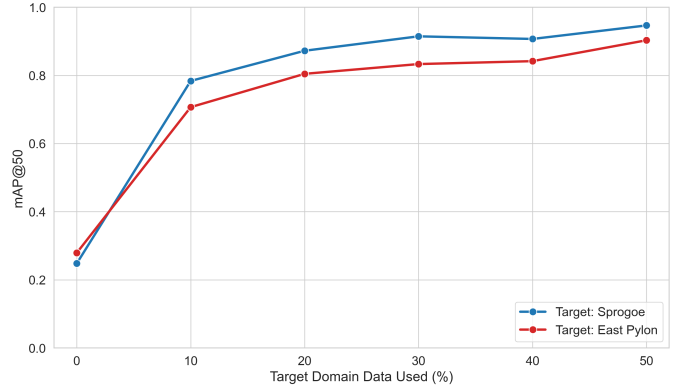


Fig. 5: Cross-camera transfer.

the other target camera. We increase the amount of target camera data used for fine-tuning progressively. The results are reported in Fig. 5. Despite observing the same maritime area, performance degrades substantially under cross-camera transfer without fine-tuning, indicating a pronounced domain shift between the two viewpoints that limits direct model transfer. However, even a small amount of fine-tuning provides enough domain exposure to rectify the gap.

IV. DISCUSSION

The baseline experiments serve as an evaluation of the dataset, revealing the inherent complexities of real-world coastal surveillance. DMD contains a complex image collection with high visual variability of one of the world’s busiest shipping lanes.

The specific stressors present in the DMD are best highlighted through the provided figures. For instance, Fig. 1 illustrates the intense specular sun glare, testing a model’s ability to distinguish vessel silhouettes against saturated backgrounds. Furthermore, Figures 7 and 8 showcase the dataset’s inclusion of military targets. The corvette is specifically designed to blend into the sea-sky horizon and is a challenging target to detect. Coupled with occlusion by the bridge (see Fig. 7), this demonstrates the high-difficulty edge cases that go beyond the commercial targets found in standard maritime datasets.

Operational transitions also play a key role in the dataset’s diversity. While primarily daytime-focused, the inclusion of low-light and post-sunset scenes, such as those in Fig. 6, challenges models to transition from hull-based detection to light-signature-based detection. This reflects the reality of continuous monitoring infrastructure where appearance changes drastically over a 24-hour cycle. Additionally, the contrast between the winter scenes on Sprogø (Fig. 2) and the clear-day side views (Fig. 9) ensures that the dataset captures seasonal atmospheric haze and a wide range of vessel scales.

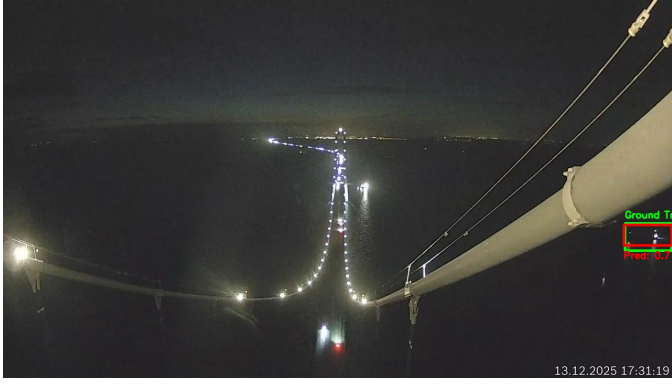


Fig. 6: Image from the East Pylon after sunset.



Fig. 7: Image from the East Pylon of an occluded camouflaged military corvette.

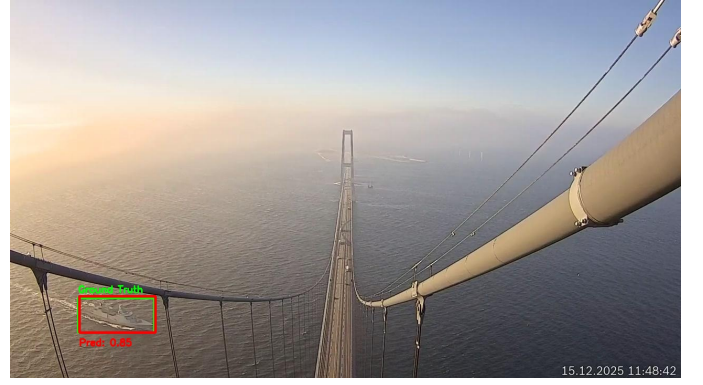


Fig. 8: Image from the East Pylon of a camouflaged military corvette.



Fig. 9: Image from Sprogø.

A. Cross-Modality Gap

The baseline experiments demonstrate the near-total failure of cross-camera generalization. Despite both sensors monitoring the exact same shipping lane, a model trained on one camera view cannot reliably detect vessels in the other, as evidenced by the dramatic performance drop in Fig. 5. This mAP_{50} decrease is not a failure of the detection framework itself but rather a highlight of the significant viewpoint diversity inherent in the DMD. The East Pylon provides an oblique, high-altitude perspective where vessels appear foreshortened and viewed from above, while the Sprogø camera provides a sea-level profile view. This generalization gap is a significant future challenge for the research community, as the DMD provides a unique opportunity to study several advanced computer vision tasks. These include Unsupervised Domain Adaptation (UDA) to bridge the gap between high-altitude and sea-level perspectives without requiring new annotations, as well as Multi-View Fusion utilizing the temporally overlapping subsets of the DMD for simultaneous multi-perspective detection. Moreover, the dataset facilitates research into maritime re-identification, using the distinct visual signatures captured by both cameras to track the same vessel as it moves through the strait.

While the Danish Maritime Dataset (DMD) is released as a visual object detection benchmark, AIS is avail-

able for the corresponding acquisition period. The data can be freely acquired from the Danish Maritime Authority at <https://www.soefartsstyrelsen.dk/sikkerhed-til-soes/sejladsinformation/ais-data>.

V. CONCLUSION

In this study, we introduced the Danish Maritime Dataset (DMD), a new publicly available benchmark for maritime object detection in high-traffic coastal environments. Collected from two distinct vantage points along the Great Belt Bridge, the dataset captures the visual complexity of real-world surveillance, including extreme scale variations, adverse weather, and challenging lighting conditions such as sun glare and low-light transitions.

Our baseline evaluations underscore the value of domain-specific pretraining, while the cross-camera experiments reveal a significant generalization gap caused by viewpoint diversity.

By providing manually annotated imagery and a unique multi-view subset, the DMD establishes a rigorous testbed for future research in maritime domain awareness. Specifically, we believe this dataset will facilitate advancements in unsupervised domain adaptation, multi-view sensor fusion, and long-term vessel re-identification.

The DMD dataset is available at: <https://zenodo.org/records/18267483>.

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